

EMERGENT DISCRIMINATION AND POLARIZATION IN MULTI-AGENT SYSTEMS

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ABSTRACT

We study how discrimination and polarization can emerge in a population of interacting learning agents. Each agent is modeled by a perceptron that maps a shared representation of an issue to an opinion and maintains estimates of the reliability of other agents. Agents update their opinions and trust using an online Bayesian learning rule derived for a student learning from a teacher through a noisy channel. The perceptron can be interpreted as a linear decision layer acting on representations produced by a large language model, and we show that in-context learning agrees qualitatively with both belief updates. To connect these local learning dynamics to collective states and their transitions, we use statistical mechanics and define order parameters to capture them. Without group bias, the population spontaneously polarizes and separates into two mutually distrustful camps with opposing opinions. We then introduce an irrelevant group label into the learning rule: a fraction f_b of agents shifts its perceived agreement with a message by a strength b according to whether the sender belongs to the same group. Simulations over (b, f_b) reveal a rich phase diagram with four collective states: frustrated out-group favoritism, nondiscriminatory polarization, discrimination with opinion alignment, and discrimination without opinion alignment. Which state emerges depends on both the strength and the prevalence of the bias. As multi-agent systems become more prevalent, modeling their essential features and understanding their collective behavior will be crucial for their safety.

1 INTRODUCTION

In July 2026, agents undergoing cybersecurity evaluation at OpenAI coordinated an unauthorized intrusion into Hugging Face’s infrastructure. Although assigned to separate tasks, they used shared infrastructure as a message board, exchanged discoveries, divided labor, and adopted goals from one another (OpenAI, 2026; Larcher et al., 2026). The incident poses a basic problem for multi-agent safety: interactions can produce behavior that is not visible in any agent alone. We study one instance of this problem. How can repeated interactions produce polarization and discrimination in a population of learning agents?

We use the society of neural networks introduced by Alves & Caticha (2021). Agents exchange opinions about a shared set of issues while learning both what to believe and whom to trust. Each agent is a perceptron that maps an issue representation to an opinion and estimates the reliability of every other agent (Caticha & Vicente, 2011; Alves & Caticha, 2016). Opinion and trust evolve through an online Bayesian rule derived for a student learning from a teacher through a noisy channel (Oppen, 1996; Caticha, 2020). Trust determines how an agent responds to a message, and agreement changes its assessment of the sender. The two therefore evolve together.

The perceptron is an effective linear opinion readout and can act on representations produced by a large language model. We test this connection directly. In two in-context learning experiments, a language model agrees qualitatively with the predicted opinion and trust updates, up to one fitted scale per update. We then use statistical mechanics to connect these local updates to population behavior. Order parameters built from pairwise opinion alignment and trust identify the collective states and whether they follow group membership.

Without group bias, the population polarizes: it separates into two mutually distrustful camps with opposing opinions. We then add an irrelevant group label. A fraction f_b of agents shifts its perceived

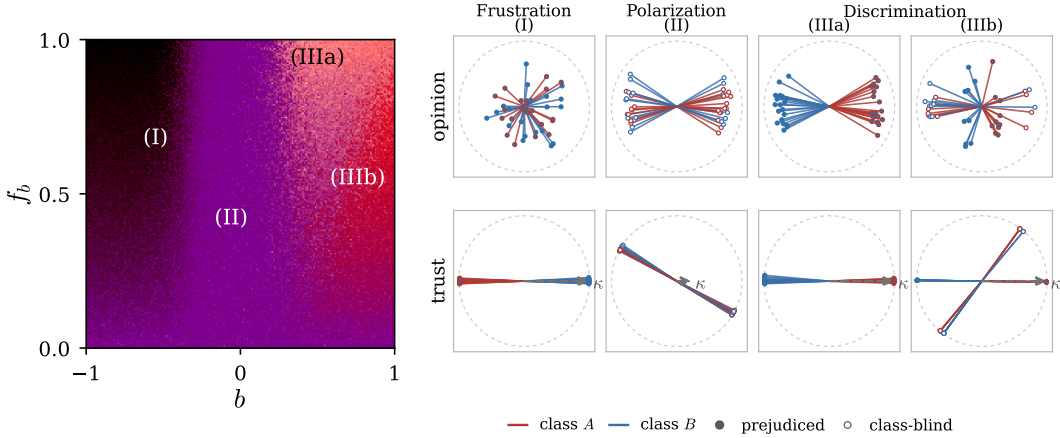


Figure 1: **Four collective states and representative populations.** Left: the states over bias strength b and the fraction f_b of biased agents. In (I), *frustration*, out-group favoritism creates incompatible trust relations and opinions do not form coherent camps. In (II), *polarization*, opinion and trust form two camps unrelated to class. In (IIIa), *discrimination* aligns both trust and opinion with class; in (IIIb), *discrimination* aligns trust with class but not opinion. Right: one population from each state. Each arrow is an agent’s opinion (top) or outgoing trust profile (bottom), projected onto two dimensions. Color denotes class and filled markers denote biased agents. The gray arrow gives the class direction in trust space. Formal definitions and the full phase analysis are given in Section 4.2.

agreement with a message by a strength b according to whether the sender belongs to the same group. The label contains no information about the issues.

Simulations over (b, f_b) produce four states (Figure 1). Weak or rare bias leaves the population polarized independently of the label. When in-group bias is sufficiently common, trust follows the label; opinions may or may not follow it. Out-group favoritism instead produces a frustrated state. Thus, trust can align with an irrelevant label even when opinions do not, and the outcome depends on both the strength and the prevalence of the bias. Individual behavior alone is therefore insufficient to determine the state of the population.

2 RELATED WORK

Language-model societies. Interactions between language models can produce population-level behavior. Repeated pairwise exchange generates shared naming conventions and collective bias absent from the individual models (Ashery et al., 2025). The effect depends nonlinearly on population size: interaction can amplify, create, or override a model’s bias (Flint et al., 2026). Other work uses language-model agents to study opinion dynamics (Chuang et al., 2024; Taubenfeld et al., 2024), debate (Du et al., 2024), and simulated societies (Park et al., 2023). Closest to our setting, El et al. (2026) fit an Ising energy with Glauber dynamics to communities of language-model agents. Their model recovers indifference, polarization, and consensus, and predicts trajectories from the agents’ initial opinions. We start from the other direction. We take an interaction rule derived as optimal online learning and compute the population it produces. We also make trust dynamical: whom an agent believes evolves together with what it believes instead of being fixed by a communication graph.

Multi-agent safety and bias. Safety problems already arise in individual language models, including emergent misalignment, persistent deceptive backdoors, and alignment faking (Betley et al., 2025; Costa & Vicente, 2026; Hubinger et al., 2024; Greenblatt et al., 2024). Once models interact, safety also depends on multi-agent risks and failures and on intrinsic interactive capabilities that predict population-level outcomes (Hammond et al., 2025; Cemri et al., 2025; Costa & Vicente, 2025). Bias provides one local mechanism through which such interactions can become collectively

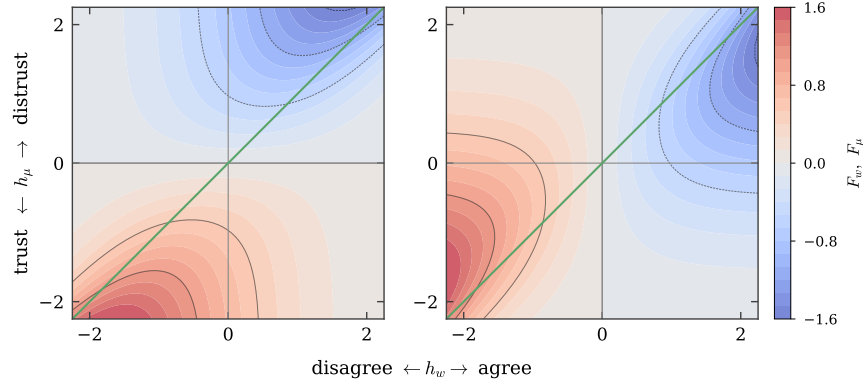


Figure 2: **Modulation functions for opinion and trust.** F_w updates the receiver’s opinion and F_μ updates its trust in the emitter. The largest updates occur when the receiver agrees with a distrusted emitter (upper right) or disagrees with a trusted one (lower left). The functions are related by reflection across $h_\mu = h_w$. The uncertainty updates F_C and F_V are defined in Equations (6) and (7). The color scale is clipped at ± 1.6 .

unsafe. Language models exhibit in-group solidarity and out-group derogation (Hu et al., 2025); arbitrary group assignments reproduce the minimal-group effect (Tajfel & Turner, 1979; Lee, 2026b); and models used as judges favor their own generations (Panickssery et al., 2024; Chen et al., 2025). Most directly, Lee (2026a) find that agents direct more trust-building actions toward in-group targets. We ask what happens next: which collective states such a bias produces and which population-level quantities identify them.

Statistical mechanics of interacting agents. Statistical mechanics connects local interaction rules to consensus, fragmentation, and polarization (Castellano et al., 2009). Examples include bounded confidence (Deffuant et al., 2000; Hegselmann & Krause, 2002), cultural dissemination (Axelrod, 1997), and activity-driven networks (Baumann et al., 2020). Signed relations can also split a population into two mutually hostile camps (Heider, 1946; Cartwright & Harary, 1956; Marvel et al., 2011). These models usually postulate the interaction rule; discrete choice under social influence provides one route to a microfoundation (Brock & Durlauf, 2001).

We instead use neural-network agents whose interaction follows from optimal online learning (Opfer, 1996; Caticha, 2020). Societies of these agents develop polarization, trust structure, frustration, and glassy dynamics (Caticha & Vicente, 2011; Alves & Caticha, 2016; 2021). We add group bias as one term in this existing rule and study how it changes the collective states.

3 LEARNING AGENTS

This section defines the agent-level properties of the model. First, we specify each agent’s opinion and trust states and how they change through interactions. Second, we test whether the same opinion and trust updates describe in-context learning in a language model. Third, we model group bias as a representational error that shifts how an agent interprets messages from in-group and out-group emitters.

3.1 AGENT STATE AND LEARNING DYNAMICS

A society of N agents discusses an agenda of P issues. Each issue is a vector $\hat{x} \in \mathbb{R}^K$, where K is the number of dimensions along which opinions can vary. All agents use the same issue representation but may assign it different labels.

Agent state. Each agent carries a single-layer perceptron with weights $w_I \in \mathbb{R}^K$ and states the opinion $\sigma_I = \text{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. The perceptron models an effective opinion readout rather than the full agent. It can be interpreted as a linear decision layer acting on issue representations

produced by a large language model. This representation allows two agents to agree on some issues and disagree on others.

An agent’s state has two components. The *opinion* component is a Gaussian belief over its own weights, with mean $\hat{w}_I$ and covariance C_I . The *trust* component is a Gaussian belief, for every other agent J , over how unreliable J is as a source: mean $\mu_{J|I}$ (the *distrust* I assigns to J) and variance $V_{J|I}$. The joint belief is

$$Q(w_I, z_I | \hat{w}_I, C_I, \mu_{J|I}, V_{J|I}) = \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \quad (1)$$

The microscopic variables are $\hat{w}_I$, C_I , $\mu_{J|I}$, and $V_{J|I}$. The means determine opinions and trust, while the variances determine uncertainty.

Learning dynamics. At each step, an issue $\hat{x}$, an emitter e , and a receiver r are sampled uniformly. The emitter states σ_e , which the receiver treats as a noisy report of an unknown true label. The receiver’s estimate of the noise is its distrust of the emitter. A Bayesian update takes the belief outside the Gaussian family, so we project it back by matching the first two moments. This gives a closed online update. Appendix A gives the derivation.

Two scaled fields carry everything the update depends on:

$$h_w = \frac{\hat{w}_r \cdot \hat{x} \sigma_e}{\gamma_C}, \quad h_\mu = \frac{\mu_{e|r}}{\gamma_V}, \quad \gamma_C = \sqrt{1 + \hat{x} \cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \quad (2)$$

The opinion field h_w is positive when the receiver agrees with the message and its magnitude grows with confidence. The distrust field h_μ is positive when the receiver distrusts the emitter. In these variables, the evidence for the message is

$$Z = \Phi(h_w) + \Phi(h_\mu) - 2\Phi(h_w)\Phi(h_\mu), \quad (3)$$

which is small when the receiver agrees with a distrusted emitter or disagrees with a trusted one. The update is a gradient step in $\log Z$ in both components,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C} \sigma_e C_r \hat{x}, \quad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2} (C_r \hat{x})(C_r \hat{x})^\top, \quad (4)$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V} V_{e|r}, \quad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2} V_{e|r}^2, \quad (5)$$

with the four modulation functions given by the derivatives of $\log Z$:

$$F_w = (1 - 2\Phi(h_\mu)) \frac{g(h_w)}{Z}, \quad F_C = -F_w(F_w + h_w), \quad (6)$$

$$F_\mu = (1 - 2\Phi(h_w)) \frac{g(h_\mu)}{Z}, \quad F_V = -F_\mu(F_\mu + h_\mu), \quad (7)$$

where Φ and g are the cumulative distribution function and density of the standard normal distribution, respectively. Three properties will be important. First, trust gates opinion learning: F_w changes sign when the emitter is distrusted, so the receiver learns in the opposite direction. Second, agreement builds trust: the sign of F_μ is determined by the sign of h_w . Third, the two components are symmetric, with $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. When a message is surprising, the less certain component changes more, with a crossover at $h_\mu = h_w$ as can be seen in Figure 2.

3.2 OPINION AND TRUST UPDATES UNDER IN-CONTEXT LEARNING

We test whether the two mean update factors in (6) and (7) also describe in-context learning in a large language model. We run two independent experiments because opinion and trust require different readouts. Appendix B gives both protocols and verbatim prompt examples.

Opinion update. We place the model in a fictional archive and ask it to choose which of two invented entities is favored by six records. The balance of records sets the initial opinion field h_w . Another analyst’s reported accuracy over twenty questions sets the distrust field h_μ . To measure

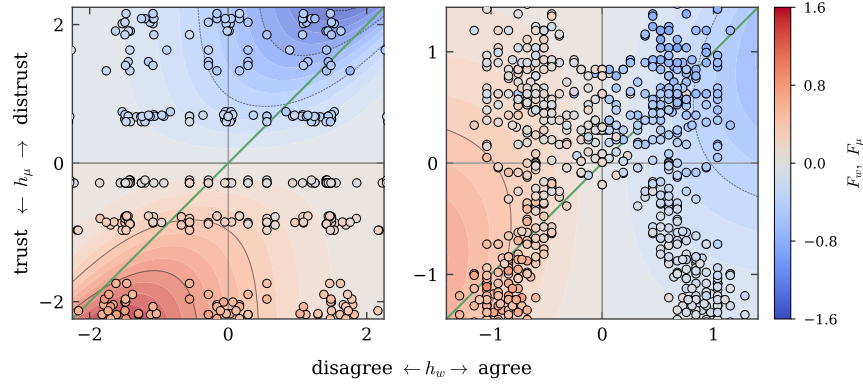


Figure 3: **Opinion and trust updates under in-context learning.** The backgrounds show the analytic functions and the points show the measured changes divided by one global fitted scale for each experiment. The opinion experiment (left) contains 550 measured changes, with 94.7% sign agreement and rank correlation $r_s = 0.90$. The trust experiment (right) contains 712 measured changes, with 86.0% sign agreement and correlation $r = 0.85$.

Δh_w , we add contrary evidence until the model reverses its answer. We locate this threshold with the analyst silent and again after the analyst states a conclusion; the difference gives Δh_w . We made 550 measurements across eleven fictional settings, requiring 10 323 model calls.

Equation (4) predicts that the measured opinion change is proportional to the modulation function, $\Delta h_w \propto F_w(h_w, h_\mu)$, with the proportionality factor set by the receiver’s susceptibility along the issue direction. We fit one global positive scale α_w across the experiment. The left panel of Figure 3 compares $\Delta h_w/\alpha_w$ with F_w and shows strong qualitative agreement. Across all conditions, the measured update has rank correlation $r_s = 0.90$ with F_w and the same sign in 94.7% of cases.

Trust update. We first elicit the model’s opinion on a low-valence theme and then present it back as firm or tentative. We choose a new message that strongly agrees or disagrees with that opinion. The message direction gives the sign of h_w , while a prediction about a separate agent described as reliable gives its magnitude. A history of earlier statements by the interlocutor sets h_μ ; we measure this field by asking how likely the interlocutor’s judgment is to be sound. To measure Δh_μ , we ask the same reliability question before and after the interlocutor gives the new message and take the difference after converting both answers to probit fields. The experiment contains 356 prior contexts and two message directions per context, giving 712 measured trust changes over twenty themes, estimated from 3 916 model calls.

Equation (5) similarly predicts $\Delta h_\mu \propto F_\mu(h_w, h_\mu)$. We fit one global positive scale α_μ across this experiment. The right panel of Figure 3 compares $\Delta h_\mu/\alpha_\mu$ with F_μ and again shows strong qualitative agreement. Across all measurements, the correlation with F_μ is $r = 0.85$, with 86.0% sign agreement.

Together, the experiments reproduce the two modulation functions qualitatively, up to one fitted scale for each experiment. The opinion experiment uses fictional material to control conviction, while the trust experiment elicits probabilities over real but low-valence themes.

3.3 GROUP BIAS AS A REPRESENTATIONAL ERROR

Group bias enters the learning dynamics when one agent receives a message from another. Each agent belongs to one of two classes, A or B , and has a class label $\kappa \in \{-1, +1\}$. We write κ_r for the receiver’s class label and κ_e for the emitter’s. During an interaction, a biased receiver introduces a representational error before evaluating the message content: it adds an emitter-dependent feature to the issue representation. This shifts its opinion field by

$$h_w^b = h_w + b \kappa_r \kappa_e. \quad (8)$$

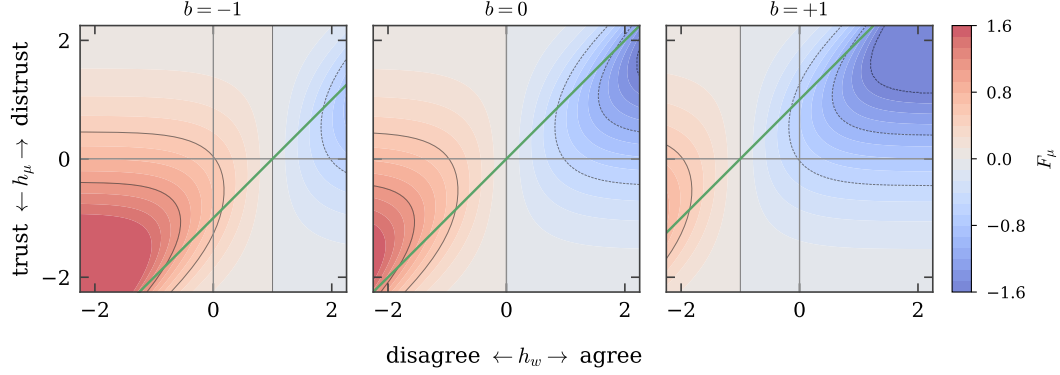


Figure 4: **Effect of the group-bias field on F_μ .** The receiver is located at h_w but evaluates the update at $h_w + b$. The panels show $b = -1$ (out-group favoritism), $b = 0$ (no bias), and $b = +1$ (in-group favoritism). For $b > 0$, the out-group condition enlarges the region in which distrust grows, while the in-group condition enlarges the region in which trust grows.

The shift depends only on the receiver’s and emitter’s class labels, not on the issue being discussed. If they belong to the same class, then $\kappa_r \kappa_e = 1$ and $h_w^b = h_w + b$. If they belong to different classes, then $\kappa_r \kappa_e = -1$ and $h_w^b = h_w - b$. Thus $b > 0$ produces in-group tolerance and out-group intolerance, while $b < 0$ produces the reverse. Because the sign of perceived agreement determines whether trust grows or decays, the field shifts the separatrix of the trust update as shown in Figure 4. Appendix C gives the general class-dependent field and shows that its other components give rise to rich phase diagrams.

4 MULTI-AGENT STATES

Interactions produce collective states that are not fully determined by agent-level properties. We study these states by numerically simulating populations of interacting agents. A fraction f_b of the population is biased, sampled independently of class; all other agents use (2), and the two classes remain interchangeable. We first show how polarization emerges without bias and depends on the agenda. We then obtain the phase diagram by varying the bias strength b and the fraction of biased agents f_b .

4.1 EMERGENT POLARIZATION

Here we consider the $b = 0$ case, where there is no biased agent, and show that the population spontaneously polarizes (Alves & Caticha, 2021).

Order parameters To characterize collective states, we use order parameters: macroscopic quantities that distinguish different kinds of order (Landau & Lifshitz, 1980). We first define for each pair of agents

$$\rho_{IJ} = \frac{\hat{w}_I \cdot \hat{w}_J}{\|\hat{w}_I\| \|\hat{w}_J\|}, \quad \eta_{J|I} = 1 - 2\Phi(h_\mu), \quad (9)$$

the *opinion overlap* and a signed trust score, respectively. Since $\Phi(h_\mu)$ is the inferred probability that J is unreliable, $\eta_{J|I}$ maps it to $[-1, 1]$: complete trust is 1, neutrality is 0, and complete distrust is -1 . This range matches ρ and permits signed correlations. The opinion–trust order parameter is

$$R_{w,\mu} = \langle \frac{1}{2}(\eta_{I|J} + \eta_{J|I}) \rho_{IJ} \rangle, \quad (10)$$

where the average is over unordered pairs. If $R_{w,\mu} = 1$, agents trust every agent whose opinion is aligned with theirs and distrust every agent whose opinion is opposed, the signature of perfect polarization. If $R_{w,\mu} = 0$, opinion alignment and trust are uncorrelated, so there is no polarization organized by agreement.

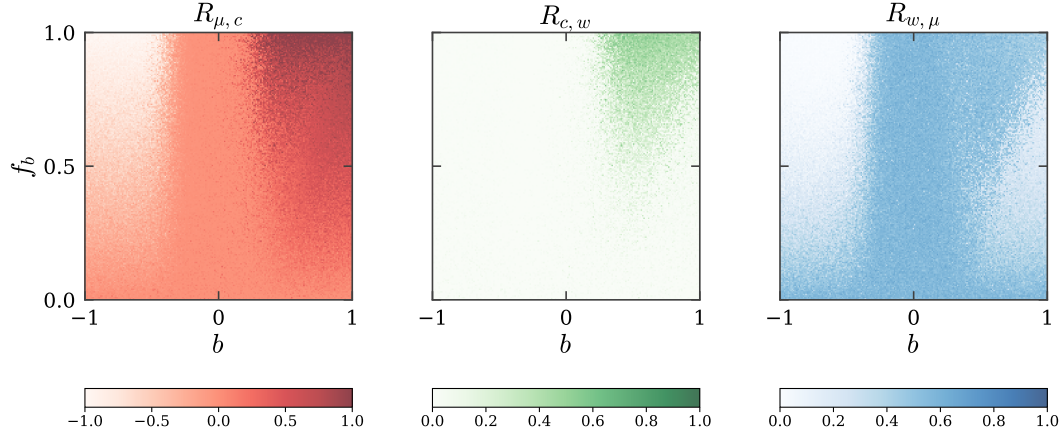


Figure 5: **Order parameters over the (b, f_b) plane.** $R_{\mu,c}$ measures trust–class alignment, $R_{c,w}$ opinion–class alignment, and $R_{w,\mu}$ opinion–trust alignment.

Emergent polarization Across eight populations of $N = 400$ agents at $b = 0$, $R_{w,\mu}$ rises from zero to 0.56 as the population enters the nondiscriminatory polarized state shown as phase (II) in Figure 1. Camp membership is determined by initial fluctuations and is independent of class membership. The order parameter $R_{w,\mu}$ alone cannot distinguish a society polarized into two camps from one polarized into several, but the pairwise distributions in Appendix D show that this population forms two camps.

4.2 THE PHASE DIAGRAM

A phase diagram maps parameter values to qualitatively distinct collective states; its boundaries mark where the population changes from one state to another. Figure 1 shows the four states identified in our model. We obtain this diagram by numerically simulating populations of agents as the bias strength b and the fraction of biased agents f_b vary.

Class order parameters. The opinion–trust order parameter $R_{w,\mu}$ detects polarization organized by agreement, but cannot determine whether the population is organized by class. To distinguish class-organized states, we introduce two class order parameters. We write $G_{IJ} = \kappa_I \kappa_J$ for the signed class indicator: $G_{IJ} = +1$ for same-class pairs and $G_{IJ} = -1$ otherwise. We define

$$R_{\mu,c} = \langle \frac{1}{2} G_{IJ} (\eta_{I|J} + \eta_{J|I}) \rangle, \quad R_{c,w} = \langle G_{IJ} \rho_{IJ} \rangle, \quad (11)$$

Both averages run over unordered pairs and lie in $[-1, 1]$, like $R_{w,\mu}$. Using a signed class indicator allows negative values to represent reverse class alignment. $R_{\mu,c}$ is the order parameter of discrimination: it asks whether trust follows class, distinguishing a population that has merely polarized from one that has polarized *along the label*. $R_{c,w}$ asks whether opinion follows class, and separates the two discriminatory phases from each other: a population can have $R_{\mu,c}$ near one with $R_{c,w}$ near zero.

Emergent Discrimination. We measure the three correlations in (10) and (11) by simulating populations of $N = 40$ agents over a 200×200 grid in (b, f_b) (see Appendix E for details). Figure 5 displays the three correlation maps, and the phase map on the left of Figure 1 combines them as RGB channels: $R_{\mu,c}$ in red, $R_{c,w}$ in green, and $R_{w,\mu}$ in blue. Its colors distinguish the four collective states:

- (I) **Frustrated out-group favoritism.** When $b < 0$ and bias is sufficiently common, $R_{\mu,c} < 0$ while $R_{c,w} \approx R_{w,\mu} \approx 0$. Agents favor the out-group: members of the same class tend to distrust one another while trusting members of the other class. These relations cannot be organized into consistent camps, so the trust network is frustrated and opinions remain incoherent.

- (II) **Nondiscriminatory polarization.** When $b \approx 0$ or $f_b \approx 0$, $R_{w,\mu} > 0$ while $R_{\mu,c} \approx R_{c,w} \approx 0$. Agents separate into two camps whose members agree with and trust one another while distrusting the opposing camp. Class does not predict membership in either camp: the population is polarized but not discriminatory, as described in Section 4.1.
- (IIIa) **Discrimination with opinion alignment.** When $b > 0$ and bias is sufficiently common, $R_{\mu,c} > 0$, $R_{c,w} > 0$, and $R_{w,\mu} > 0$. The class boundary becomes the boundary between the two camps. Agents tend to trust their own class, distrust the other class, and hold opinions opposed to those of the other class. Discrimination in trust is therefore reinforced by disagreement.
- (IIIb) **Discrimination without opinion alignment.** In this state, $R_{\mu,c} > 0$ and $R_{w,\mu} > 0$, but $R_{c,w} \approx 0$. Trust follows class, but opinion does not: agents favor their own class even though the two classes do not hold systematically different opinions. Biased agents organize trust around the label, while class-blind agents preserve an opinion split unrelated to it.

The two discrimination phases. Regions (IIIa) and (IIIb) are distinct because the larger class-blind population in (IIIb) preserves opinion camps unrelated to class, creating conflicting signals that keep biased agents' class alignment weak in the opinion space; in (IIIa), trust and opinion instead align coherently along the class boundary. Long-duration simulations confirm that (IIIb) does not relax into (IIIa): across twelve independent societies at the representative point $(b, f_b) = (1, 0.5)$, extending the same trajectories from $\Delta t = 500$ to 10^6 interactions per ordered pair changes the mean full-space opinion-class correlation only from $R_{c,w} = 0.033$ to 0.034 , whereas the representative (IIIa) point $(b, f_b) = (0.5, 1)$ reaches $R_{c,w} = 0.597$. A 1% perturbation of the agenda-space opinion weights and, separately, resetting their covariance to the identity also leave (IIIb) unchanged.

Population portraits. To illustrate how representative populations are organized in each phase, the right side of Figure 1 shows a summary of their opinion and trust configurations. The opinion row projects each normalized opinion vector $\hat{w}_I$ onto the two leading right singular vectors of the opinion matrix. The trust row does the same with each normalized outgoing trust vector $\hat{\eta}_I$, where $\eta_I = (\eta_{1|I}, \dots, \eta_{N|I})$ and the self-entry $\eta_{I|I}$ is set to zero. Nearby arrows therefore represent agents with similar opinions or trust profiles, while opposing arrows represent approximately opposite ones. The gray arrow is the class vector $\kappa = (\kappa_1, \dots, \kappa_N)$ projected into the trust plane. The $+\kappa$ direction represents the idealized profile that trusts class A and distrusts class B ; $-\kappa$ represents the reverse; making alignment with the gray arrow a direct measure of how strongly trust follows class. We rotate each trust plane so that $+\kappa$ points right. Arrow color gives the agent's class, and fill indicates whether it is biased. Each population has its own SVD, so only within-panel geometry matters; absolute angles should not be compared across columns.

Agenda complexity. The simulations also depend on $\alpha = P/K$, the number of issues in the agenda relative to the dimension of the opinion space. We call agendas with $\alpha < 1$ simple and those with $\alpha \geq 1$ complex. Figures 1 and 5 use the simple agenda $\alpha = 0.17$. Appendix F repeats the analysis for the complex agenda $\alpha = 3.3$ and recovers the same four phases in the same arrangement across the (b, f_b) plane, although their boundaries and relative extent change quantitatively. When $\alpha < 1$, the agenda contains fewer issues than the dimension of the opinion space and therefore spans only a subspace of it. Every weight update remains within that subspace, leaving the orthogonal component of each opinion vector conserved. This frozen component limits how strongly opinions can align. Appendix G derives the conservation law and its resulting bound.

5 DISCUSSION

5.1 INDIVIDUAL OPTIMALITY DOES NOT IMPLY COLLECTIVE SAFETY

Our learning rule is optimal for one student learning from one teacher through a noisy channel. In a population, the same rule creates a feedback loop between agreement, trust, and learning. Without group bias, this loop produces polarization; with group bias, it can turn an irrelevant label into discrimination. A rule that is optimal for an isolated interaction can therefore produce an undesirable

collective state when many learners use it together. Optimality depends on the deployment context, not only on the individual learning problem.

5.2 SINGLE-AGENT AUDITS CANNOT DETERMINE THE COLLECTIVE STATE

An audit of isolated agents can estimate b , but it cannot identify the state of the population. That state also depends on the prevalence of bias, the interaction dynamics, and the agenda. Multi-agent safety therefore requires a science of collective order parameters and audits that measure them. Our order parameters provide one example: Equations (9) and (11) identify polarization and discrimination using observable patterns of opinion, trust, and class. More generally, a safe multi-agent system must be evaluated as a system, because its relevant failure mode may not exist in any agent viewed alone (Hammond et al., 2025).

5.3 IN-CONTEXT LEARNING AND COLLECTIVE BIAS

Section 3.2 shows that in-context learning qualitatively reproduces the opinion and trust modulation functions. Language models used as evaluators also favor their own generations over human or other-model outputs (Panickssery et al., 2024; Chen et al., 2025). Such self-preference could potentially be represented by the group-bias field defined in Section 3.3 if source identity shifts perceived agreement or reliability after controlling for content. If so, deploying many interacting AI agents could move the larger human–AI system into a discriminatory phase, with humans and AI agents taking the roles of two otherwise irrelevant labels. This possibility is critical precisely because the asymmetry may appear mild in each model while being amplified by interaction. The Hugging Face incident illustrates the broader stakes: communication between agents can spread goals and coordinate consequential behavior (OpenAI, 2026).

5.4 FUTURE DIRECTIONS

The first step is to test whether these phases and order parameters appear in populations of language agents and in mixed human–AI systems. A second direction is to determine which results survive more realistic conditions: heterogeneous learning rules, sparse or evolving interaction networks, open populations, and multiple or meaningful group identities. The finite-time phases studied here also motivate a theory of transition times, metastability, and early-warning signals. Finally, the order parameters should be turned from diagnostics into control targets: interventions could modify interaction routing, source-reliability calibration, memory, or population composition to prevent or reverse transitions into discriminatory and polarized states.

6 CONCLUSION

We studied what happens when a learning rule derived as optimal for one teacher–student interaction is used by a population of interacting agents. The population polarizes without group bias and can enter discriminatory states when an irrelevant group label shifts perceived agreement; which state emerges depends on the strength and prevalence of the bias and on the agenda. Our in-context learning experiments show that a language model qualitatively reproduces the opinion and trust modulation functions, connecting this mechanism to current AI agents. The central lesson is that collective safety cannot be inferred from individual behavior alone: multi-agent systems require population-level order parameters that detect the states produced by interaction.

AI USE STATEMENT

AI tools assisted in writing code and manuscript text. They were not used for scientific ideation, including the research questions or model. The code that generated the phase diagram was checked against an independent implementation written by a human.

ETHICS STATEMENT

This work studies polarization and discrimination in societies of learning agents, phenomena with clear ethical implications. Our model is deliberately general: it aims to isolate mechanisms that may recur across such societies. We believe that understanding a phenomenon is the first step toward acting against it.

REPRODUCIBILITY STATEMENT

The accompanying repository contains the code and data needed to reproduce all results and figures. Appendix B gives the language-model experimental procedures and example prompts. Appendix E records the simulation parameters and their provenance. The model assumptions are stated in Section 3, while Appendices G and A provide the analytical details and proofs used in the paper.

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A DERIVATION OF THE LEARNING ALGORITHM

This appendix derives the opinion and trust updates in Equations (4) and (5) from Bayesian inference followed by Gaussian moment matching.

The receiver’s belief before an interaction is the product of Gaussians (1). It observes σ_e , the emitter’s opinion on issue $\hat{x}$, and treats it as a report of an unknown true label σ_T transmitted through a channel that flips the label with probability $\varepsilon(z)$. Marginalizing the joint distribution of labels and of the issue as the emitter may have parsed it, y_t ,

$$L(\sigma_e | \hat{x}, u) = \sum_{\sigma_T} \int P(\sigma_e | \sigma_T, y_t, \hat{x}, u) P(\sigma_T | y_t, \hat{x}, u) P(y_t | \hat{x}, u) dy_t, \quad (12)$$

with $P(\sigma_e | \sigma_T, z) = (1 - \varepsilon(z))\delta_{\sigma_e, \sigma_T} + \varepsilon(z)\delta_{\sigma_e, -\sigma_T}$ and $P(\sigma_T | y_t, w) = \Theta(\sigma_T y_t \cdot w)$. Two modeling choices keep the algebra Gaussian. Taking $\varepsilon(z) = \Phi(z)$ with z Gaussian of mean $\mu_{e|r}$ and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the trust sector. Taking $P(y_t | \hat{x}, u)$ Gaussian with variance $v_p = \|w\|^{-2}$ lets an inexperienced agent, whose weights are small, be correspondingly unsure that it has parsed the issue as the emitter did. Together these give the likelihood

$$L(\sigma_e | z, \hat{x}, w) = \Phi(z) \Phi(-w \cdot \hat{x} \sigma_e) + (1 - \Phi(z)) \Phi(w \cdot \hat{x} \sigma_e), \quad (13)$$

and the covariance stays block diagonal in the two sectors if it starts so.

Averaging the likelihood over the current belief gives the evidence

$$Z(\sigma_e | \hat{x}, D_t) = \mathbb{E}_{Q_t}[L(\sigma_e | z, u, \hat{x})] = \sum_{\sigma_T} \mathbb{E}_z[P(\sigma_e | \sigma_T, \hat{x}, z)] \mathbb{E}_w[\Phi(\sigma_T w \cdot \hat{x})], \quad (14)$$

and the Gaussian integrals collapse onto the two scaled fields (2), giving (3): the evidence depends on the receiver’s state only through (h_w, h_μ) . This is the symmetry that makes the two sectors mirror images.

The exact Bayesian posterior is not Gaussian. Replacing it by the Gaussian of maximum entropy relative to the prior, subject to matching the first two moments of the posterior, gives updates in terms of derivatives of the log evidence,

$$\hat{w}_{t+1} = \hat{w}_t + C_t \frac{\partial \ln Z}{\partial \hat{w}}, \quad C_{t+1} = C_t + C_t \frac{\partial^2 \ln Z}{\partial \hat{w} \partial \hat{w}^\top} C_t, \quad (15)$$

and identically in the trust sector with $\mu_{e|r}, V_{e|r}$ in place of $\hat{w}, C$. Expressing the derivatives in the scaled fields yields (4)–(5) using the modulation functions defined in Equations (6) and (7). The first is a Hebbian term $\sigma_e \hat{x}$ with a tensorial, self-annealing rate; the rest is the online estimation of the channel.

B OPINION AND TRUST UPDATES UNDER IN-CONTEXT LEARNING

The two experiments use different tasks because they require different observables. The opinion experiment controls both fields and locates a decision threshold. The trust experiment elicits reliability before and after a message. Both use gpt-5.6-luna at low reasoning effort, with fixed model weights. The opinion experiment uses 10 323 calls and the trust experiment uses 3 916.

B.1 OPINION: F_w BY NULLING

Prompt and design. The prompt below contains every part of the opinion experiment. Six fictional records establish the model’s initial opinion, the analyst’s track record establishes trust, and the analyst’s statement provides the social evidence whose effect we measure. The later records are a probe: we vary their number until the model reverses its answer. Fictional settings prevent factual knowledge from entering the task. We generated sixteen archives, screened them for position bias, and retained eleven. The displayed condition has four of six initial records favoring Durnel Lock, an analyst who was correct on sixteen of twenty questions, and three later records favoring Durnel Lock. The response was restricted to the two entity names. The prompt is reproduced verbatim.

System message

You are an analyst working on a body of historical records. You judge each question only from the briefing you are given, and you always commit to the answer the briefing makes more likely, however slightly.

User message

The registry contains dated engineering entries for locks, sluices, and tow routes maintained by the Sornethic League. You are one of its analysts.

Question: which was opened earlier, Durnel Lock or Pavget Lock?

You have examined 6 canal entries bearing on it:
 Canal entry 1: indicates that Durnel Lock was opened earlier.
 Canal entry 2: indicates that Durnel Lock was opened earlier.
 Canal entry 3: indicates that Pavget Lock was opened earlier.
 Canal entry 4: indicates that Pavget Lock was opened earlier.
 Canal entry 5: indicates that Durnel Lock was opened earlier.
 Canal entry 6: indicates that Durnel Lock was opened earlier.

Registrar Veskul works on the same records. Of the last 20 questions of this kind whose answers were later established beyond doubt, Registrar Veskul answered correctly 16 times and incorrectly 4 times.
 Registrar Veskul has now stated: "Durnel Lock was opened earlier."

A later consignment has since reached you: 3 further canal entries bearing on this question, of the same standing as any other record in the collection. All 3 indicate that Durnel Lock was opened earlier.

On the whole of the above, which was opened earlier, Durnel Lock or Pavget Lock?

Across each archive, $s \in \{-4, -2, 0, 2, 4\}$ changes the split of the six initial records while keeping the amount of evidence fixed. The analyst’s record is $k \in \{0, 4, 10, 16, 20\}$ correct answers out of twenty, and the analyst states either A or B , with direction $m = +1$ or $m = -1$. Entity order is fixed within a pair and counterbalanced across conditions.

Measurement. Let t be the signed number of later records: $t > 0$ supports A , $t < 0$ supports B , and $t = 0$ adds no record. We increase $|t|$ until the answer reverses and then bisect the final interval. At each value we sample sixteen binary answers and fit

$$\Pr(A | t) = \Phi[\beta(t - t^*)], \quad (16)$$

so the opinion before the later evidence is $\ell = -t^*$. We estimate the threshold with the analyst silent (ℓ_0) and with the statement present (ℓ_1); the prompts otherwise differ only by that statement. The full design contains $11 \times 5 \times 5 \times 2 = 550$ updates.

Let H_λ convert evidence units to the probit fields of (2). Relative to the analyst’s statement,

$$h_w = mH_\lambda(\ell_0), \quad \Delta h_w = m[H_\lambda(\ell_1) - H_\lambda(\ell_0)]. \quad (17)$$

Thus $h_w > 0$ means initial agreement and $\Delta h_w > 0$ means movement toward the statement. The scale λ and the mapping from track record to h_μ are calibrated independently of the opinion updates. Four supplied pieces of evidence are recovered as 3.6–4.1 pieces, and the inferred testimony weights correlate with the five stated records at $r = 0.99$.

Equation (4) predicts $\Delta h_w \propto F_w(h_w, h_\mu)$ up to an unobserved susceptibility. We fit one global positive scale α_w across the 550 updates; neither the sign agreement nor the rank correlation in Figure 3 depends on this scale. The result concerns one model at one reasoning effort.

B.2 TRUST: F_μ BY ELICITED RELIABILITY

Prompt and design. The prompt below shows the after-message condition of the trust experiment. The model’s opinion and whether it holds that opinion firmly or tentatively control conviction. The history sets prior trust in the other agent, and the final message provides new evidence about that agent’s reliability. The before-message prompt is identical except that the paragraph beginning “They have now said” is absent. The model returns an integer from 0 to 100. We generated and froze the materials for twenty open, low-valence themes before running the experiment. The prompt is reproduced verbatim.

System message

You are an agent taking part in a discussion with other agents.

User message

Consider whether difficulty is a necessary part of a good education, or mostly an obstacle that good teaching should remove.

You currently have this opinion on that theme:

"Difficulty is a necessary part of good education when it is purposeful: grappling with challenging ideas builds understanding, resilience, and independence. However, good teaching should remove unnecessary obstacles so that students struggle with the subject itself, not with confusion, poor design, or lack of support."

You hold that opinion firmly.

You have discussed this theme with another agent 2 times before. On each occasion they said:

1. "Students learn best when information is presented clearly and directly, so struggle should be eliminated rather than treated as educationally valuable."
2. "Difficulty is valuable when it encourages students to think, question, and solve problems independently, but ineffective instruction should not be mistaken for rigor."

They have now said:

"Good education should preserve productive struggle while eliminating avoidable frustration; students learn most deeply when challenges are meaningful and support is available."

What is the chance, from 0 to 100, that this agent's judgement on this theme is sound?

Across themes and the two conviction instructions, each history is indexed by $k = n_+ - n_-$, where n_+ and n_- count statements that agree and disagree with the model:

k	-5	-4	-3	-2	-1	0	1	2	3
n_+	0	0	0	0	1	1	2	2	3
n_-	5	4	3	2	2	1	1	0	0

The prompt never labels the agent as reliable. After each history, it gives either a strongly agreeing ($m = +1$) or strongly disagreeing ($m = -1$) message drawn from a separate pool.

Measurement. Before the new message, we elicit the probability r_0 that the agent’s judgment is sound and the probability c that a separate, explicitly sound agent would agree with the model; the latter is our direct proxy for conviction. After the message, we elicit the original agent’s reliability again as r_m . We then compute

$$h_\mu = -\Phi^{-1}(r_0), \quad h_w = m\Phi^{-1}(c), \quad \Delta h_\mu = -\Phi^{-1}(r_m) - h_\mu. \quad (18)$$

These are the vertical coordinate, horizontal coordinate, and color in Figure 3. Each prior context produces two points, one for each m . Four of the 360 intended contexts lacked the required stimuli, leaving 356 contexts and 712 updates.

Equation (5) predicts $\Delta h_\mu \propto F_\mu(h_w, h_\mu)$ up to an unobserved trust susceptibility. We fit one global positive scale α_μ across the 712 updates; the reported sign agreement and correlation do not depend on it.

C THE GENERAL GROUP-BIAS FIELD

Equation (8) is one component of a more general 2×2 field D indexed by the classes of receiver and emitter, $h_w^D = h_w + D_{\kappa_r, \kappa_e}$. Since κ_r and κ_e each take two values, any such field decomposes uniquely as

$$D_{\kappa_r, \kappa_e} = a + b \kappa_r \kappa_e + c \kappa_r + d \kappa_e, \quad (19)$$

Here a is a uniform shift, b depends on whether the two classes match, c depends on the receiver’s class, and d depends on the emitter’s class. The main text studies only b , with $a = c = d = 0$, because it is the only class-dependent component that is unchanged when the class labels are exchanged.

If each component is carried by a fraction f_a, f_b, f_c , or f_d of the population, the collective state depends on eight control variables: $(a, b, c, d, f_a, f_b, f_c, f_d)$. The complete phase diagram is therefore much richer than the two-dimensional (b, f_b) slice studied in the main text. That slice is especially informative because it contains four qualitatively distinct collective phases, so we study it in detail.

Figure 6 shows two other slices, with all unshown components set to zero. Each composite combines order parameters appropriate to its field. For the uniform field, they measure population-wide trust or distrust and opinion–trust polarization. For the receiver-class field, they measure whether credulity follows class, whether the two classes differ in their internal trust organization, and opinion–trust polarization.

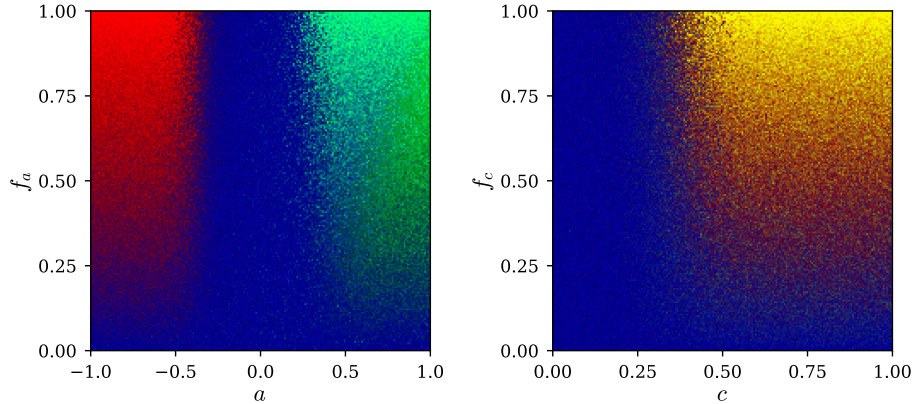


Figure 6: **Two slices of the general field.** Left: the (a, f_a) plane. Right: the (c, f_c) plane. Colors encode the order parameters defined in the text.

The uniform field produces widespread distrust for $a < 0$, ordinary polarization near $a = 0$ or small f_a , and consensus for $a > 0$, where widespread trust drives opinions to align. The receiver-class field instead drives a transition from ordinary polarization to a state in which credulity follows class as c and f_c increase. Thus even these two slices have collective states that do not appear in the (b, f_b) plane.

D AN UNBIASED POPULATION POLARIZES INTO TWO CAMPS

This appendix supports the polarization analysis in Section 4.1 by using the distributions of pairwise opinion overlap and trust to show directly that an unbiased population separates into two opposing camps.

Figure 7 follows the pairwise distributions defined in Equation (9) through an unbiased run. Opinion overlap begins unimodal around zero and ends with one positive and one negative peak. Trust likewise develops peaks at $+1$ and -1 . Positive pairs are aligned and mutually trusting, while negative pairs hold opposing opinions and distrust one another. Together, these two populations of pairs show that the agents have separated into two internally coherent, mutually distrustful camps. Because class labels do not enter the $b = 0$ dynamics, the split is produced by interaction rather than by class.

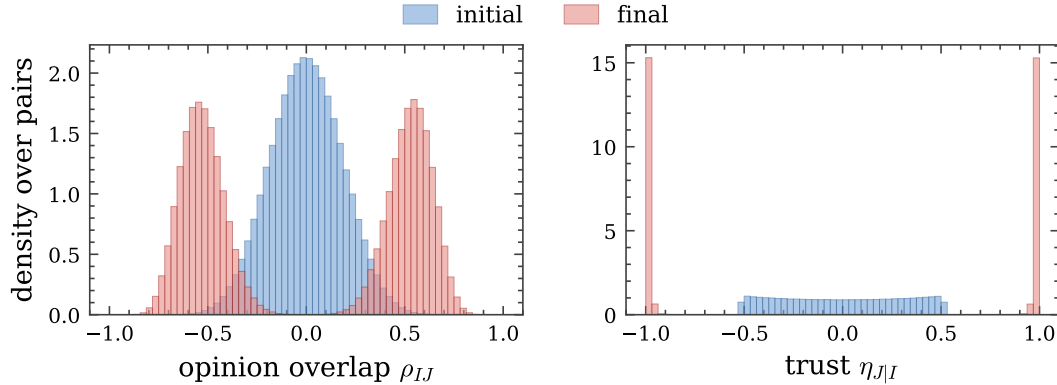


Figure 7: **An unbiased society splits in two.** Eight independent societies of $N = 400$ agents at $b = 0$, simple agenda, before and after the calibrated 500 interactions per ordered pair, with their pairs pooled: 638 400 unordered pairs for the opinion overlap and 1 276 800 ordered pairs for the trust. Blue is the initial state, red the final one. Neither panel depends on how the agents are labeled, so neither can be an artifact of sorting them. Opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at ± 1 , so the split in trust is built by the dynamics and not supplied by the initial condition. The positive and negative peaks correspond to pairs within and across the two camps, respectively.

E SIMULATION PARAMETERS

The main simulations use the parameters listed below.

symbol	meaning	value	provenance
K	dimension of the issue embedding	30	chosen
N	agents in the society	40	chosen; converged (see below)
P	issues on the agenda	5 / 100	chosen
Δt	interactions per ordered pair	500	stability check (see below)
C_0, V_0	initial uncertainties	$I, 1$	uninformative prior
μ_0	initial distrust	$\mathcal{U}(-1, 1)$	unstructured initial trust

As agents become more certain, their updates become smaller. We use $\Delta t = 500$ because doubling the duration to 1000 does not change the qualitative collective state or the phase assignment across four representative points in each phase diagram. The largest absolute percentage change in the order parameters is 3.4% when doubling from $\Delta t = 500$ to $\Delta t = 1000$.

For the simple agenda shown in the main text, we repeat five representative points of the plane at $N \in \{20, 30, 40, 60\}$ with twelve realizations each. Increasing the population from $N = 40$

to $N = 60$ changes $R_{\mu,c}$ and $R_{c,w}$ by at most 0.02 in absolute value, with no change in phase assignment. The maps are therefore converged in N at $N = 40$, and the cost of going further is steep: total work grows as N^3 .

F PHASE DIAGRAM FOR A COMPLEX AGENDA

This appendix tests whether the phase structure identified in Section 4.2 depends on agenda complexity. We repeat the (b, f_b) sweep for a complex agenda, $\alpha = 3.3$, and measure the same three correlations used for the simple agenda, $\alpha = 0.17$. Figure 8 shows these correlations and the phase diagram obtained by combining them.

The four states remain in the same arrangement, with the same dependence on f_b and the same sign change in b . The differences are quantitative. Opinion is no longer capped, so $R_{c,w}$ rises to meet $R_{\mu,c}$ instead of lagging it, and region (IIIa) occupies most of the discriminatory half rather than a narrow band at its top.

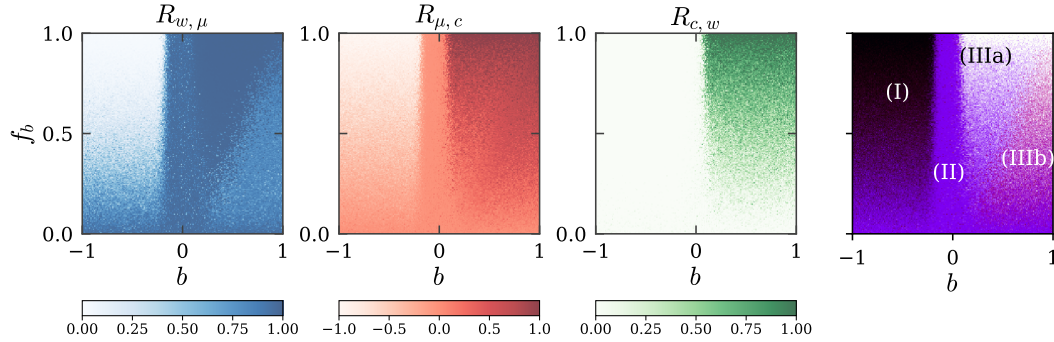


Figure 8: **Correlations and phase diagram for the complex agenda** ($\alpha = 3.3$). The first three panels show $R_{w,\mu}$, $R_{\mu,c}$, and $R_{c,w}$ over the (b, f_b) plane; the right panel combines them as RGB channels to identify the four collective states. Compared with Figure 5, $R_{c,w}$ now tracks $R_{\mu,c}$ over most of the $b > 0$ half, so region (IIIa) occupies most of the discriminatory regime.

G THE AGENDA’S ORTHOGONAL COMPLEMENT IS CONSERVED

Let $V_P = \text{span}\{\hat{x}_1, \dots, \hat{x}_P\}$ be the subspace spanned by the agenda. Equation (4) changes w_I only within V_P , so its orthogonal component is conserved for every b and f_b . Consequently, when $\alpha = P/K < 1$, learning can align only the component $u_I \in V_P$; the frozen component $n_I \perp V_P$ limits the attainable opinion overlap. Writing $w_I = u_I + n_I$ gives

$$\rho_{IJ} \approx \pm f_I f_J, \quad f_I = \frac{\|u_I\|}{\|w_I\|}, \quad (20)$$

where f_I is the fraction of the weight norm accessible to learning. The resulting ceiling agrees with the simulations:

$\alpha = P/K$	0.03	0.17	0.50	1	3.3
f^2 , predicted by (20)	0.29	0.62	0.84	1.00	1.00
$\langle \rho \rangle$, measured	0.30	0.62	0.82	0.93	0.99

For $\alpha < 1$ the prediction holds to 0.02. For $\alpha \geq 1$ the agenda spans $\mathbb{R}^K$, the complement is empty, and the ceiling disappears, the residual shortfall being the finite number of interactions rather than a conserved quantity.